

Natalie Ovcarov

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EDUCATION

New York University, Courant Institute of Mathematical Sciences | Sep 2021 - May 2025

Bachelor's of Arts, [Computer Science](#), Minor: [Web Development and Applications](#)

- Study Abroad: NYU Madrid (Fall 2024)
- Dean's List (Fall 2024) - Coursework: Software Engineering, Operating Systems, Data management and Analysis, Research.
- Graduate Course - Big Data Science (Spring 2025): Principal Component Analysis (PCA), Singular Value Decomposition (SVD), Data Preprocessing, Data Engineering, Handling Imbalanced Datasets.

WORK EXPERIENCE

Wordpress Engineer Intern | Classifiedy - Remote | February 2025 - Present

- Maintaining and enhancing the OSclass Marketplace, a CMS-based platform, ensuring performance for global users
- Developing WordPress and OSclass themes/plugins by utilizing principles in PHP and JavaScript to improve functionality
- Thrive in a fast-paced, Agile environment, collaborating in sprint planning, code reviews, and feature development

Computer Science Tutor | New York University – New York City, New York | January 2025 - May 2025

- Providing in-person and online tutoring to 5-6 students daily, guiding them through core Java concepts like object-oriented programming (OOP), recursion, inheritance, arrays, and sorting algorithms
- Debugging Java programs in Eclipse, teaching breakpoints, step-through execution, and watch variables to resolve errors
- Enhancing problem-solving and algorithmic thinking, guiding students to write efficient, structured, and maintainable code

Information Technology Intern | Union of Associations of Pensioners of Macedonia – Skopje, Macedonia | June 2023 - Aug 2023

- Provided IT support to 10+ staff members and 50+ association members during the internship
- Led the IT setup for the Annual General Meeting with 100+ attendees including the President of Macedonia, ensuring stable internet, projector functionality, microphone setup, and video conferencing for remote participants

RESEARCH

What Do Our Students Want? | Jan 2024 - Present

Supervisor: Prof. Anasse Bari, Head of AI and Predictive Analytics Research Group at NYU

- Designed and distributed the survey using [Qualtrics](#), gathering 700+ responses from NYU students since January 2024
- Currently analyzing survey responses and drafting a research paper for publication, using text preprocessing techniques to prepare data for NLP analysis with the NYU AI Lab, led by Prof. Anasse Bari

PROJECTS

Portfolio - <https://nataliovcharov.github.io/> (for additional projects)

Team Project (In Progress) - HydroForest | [GitHub Repository](#)

- Structured the Product Backlog in GitHub Issues, defining user stories, milestones, and task boards for sprint management
- Designed the [app map](#) and [Figma prototype](#), outlining user flows, structuring navigation flow and screen relationships
- Currently building the React.js front-end, implementing JSX components, and adding interactive UI elements

Team Project - WEARther | [GitHub Repository](#) | View demo here: <https://nataliovcharov.github.io/>

- Developed a full-stack web application integrating real-time weather data with location-specific outfit recommendations
- Built as a Dockerized system, using Flask (backend), MongoDB (database), and Python-based services
- Integrated the OpenWeather API to fetch real-time weather data and provide location-based recommendations
- Designed CI/CD pipelines via GitHub Actions, automating testing, container builds, and DigitalOcean deployments

Team Project - Voice Journal | [GitHub Repository](#) | View demo here: <https://nataliovcharov.github.io/>

- Developed a journaling web app that transcribes speech into text and applies sentiment analysis to track mood trends
- Built with Flask (backend), MongoDB (database), and a machine learning client, leveraging Google Speech-to-Text for transcription and TextBlob for NLP-based sentiment classification
- Implemented Flask-Login for user authentication and session management ensuring secure access to personal journal entries
- Followed Agile practices, utilizing user stories, daily standups, task boards, and pair programming to enhance collaboration

SKILLS

Agile methodologies (Scrum), Sprint Planning, Standups, Task Boards, Git & Github, GitHub Actions, Docker, DigitalOcean, MongoDB, Flask, RESTful APIs, NLP (TextBlob, Google Speech-to-Text), Express.js (basic), React (basic), Python, Java, JavaScript (basic), C (basic), Macedonian (fluent), English (fluent), Ability to quickly learn new frameworks, tools, and languages as needed.